

INTEL (IIT Kharagpur Placement, 2022)

Tuesday, October 11, 2022 11:13 PM

TEST STRUCTURE

Groups	Type of Questions	No. of Questions	Duration
Aptitude	ReferenceToContext, MCQ	15	20 mins
Technical	MCQ	20	30 mins
Coding	Coding	2	40 mins

How many pairs of letters are there in the word '**POSTMASTER**' which has as many letters between them (from both sides) in the word as in the alphabets from A to Z?

A container has three liquids A, B and C in the volume ratio 2 : 3 : 4. The volume of the container is 90 L. 30 L of the solution is taken out and then replaced with liquid A. 45 L of the solution so formed is taken out and is then replaced with liquid C. Find the ratio of the volumes so formed finally.

Santhosh and Sameer are running around a circular track of diameter 280 metres. Santhosh can go round the track once in 110 seconds and Sameer is faster than Santhosh by 50%. They started simultaneously towards each other starting from two diametrically opposite points A and B. If D is the point between A and B at which they meet for the first time, after how much time from the start do they meet at D for the fifth time?

QUESTION: **36**GROUP: **Coding**SECTION: **Language Coding 1**

Mark(s): 10

Jerome is an IT professional engineer. He has an old model configured chip in his house, which is not supported in his new laptop. He wants to re-configure it so that it can be used. To re-configure it, he has done many experiments. At last, finally he got a solution. The solution is, in the old chip there are two codes(numbers) written on it X and Y. He has to replace X with a new reconfigured code M. M must be the least number that must include X as the leading digits and it is a multiple of Y

Can you help Jerome with a program that accepts X and Y and prints new reconfigured code M of the chip.

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings anywhere in the program, as these contribute to the standard output and test cases will fail.

Constraints:

i. $X, Y \geq 1$

Input Format:

The input contains two codes X and Y of the old chip separated by a single white space.

Output Format:

The output must consist of new reconfigured code M of the chip.

Sample Input 1:

33 11

Sample Output 1:

33

Explanation 1:

Old chip two codes, X and Y: 33 and 11.

Here, by the following method of re-configuration, possibilities can be M: 33, 330, 331, 332, and so on.

Here, considering $M = 33$, its leading digit(s) is X (33) and it is a multiple of Y(11).

Hence, 33 is the new reconfigured code of the chip.

Sample Input 2:

50 3

Sample Output 2:

501

Explanation 2:

Old chip two codes, X and Y: 50 and 3

Here, by the following method of re-configuration, possibilities can be M: 50, 500, 501, 502, and so on.

Here, considering $M = 501$, its leading digit(s) are X(50) and is also multiple of Y(3).

Hence, 501 is the new reconfigured code of the chip.

QUESTION: 37GROUP: CodingSECTION: Language Coding 2

Mark(s): 10

Bob is a studious boy. He is very interested in solving difficult mathematical problems in no time. By seeing his talent, Bob's teacher gave him a problem and asked him to

solve it as soon as possible.

The problem was, Bob has been given an arithmetic expression A , the integer P , and M and he has to find the smallest non-negative value of x in expression A such that the remainder of dividing A by M equals P . Consider that solution do exist for the problems provided to him.

Therefore, if the principles of distribution are applied to expression A (Formally, the expression A is a polynomial of the first degree in variable x).

Can you help Bob with a program to solve the problem in a minimum amount of time.

Input Format:

The first line of input contains the expression A .

The second line of input contains two integers P i M .

The arithmetic expression A will only consist of characters $, -, *, (,), x$, and digits from 0 to 9. The brackets will always be paired, the operators $, -$ and $*$ will always be applied to exactly two values (there will not be an expression (-5) or $(4 -5)$) and all multiplications will be explicit (there will not be an expression $4(5)$ or $2(x)$).

Output Format:

The first and only line of output must contain the smallest non-negative value of variable x .

Constraints:

i. A ($1 \leq |A| \leq 105$)

ii. P ($0 \leq P \leq M - 1$) i M ($1 \leq M \leq 106$)

Sample Input 1:

$4+x+2$

8 9

Sample Output 1:

2

Explanation 1:

Arithmetic expression, A : $4+x+2$

P : 8 M : 9

Here, by solving the given equation, we will get:

If $x = 0$, then the remainder of dividing 4×2 with 9 is 6, which is not equal to P .

If $x = 1$, then the remainder of division 4×2 with 9 is 7, which is not equal to P .

If $x = 2$, then the remainder of division 4×2 with 9 is 8, which is equal to P .

Hence, as an output, it will print 2 as the smallest value of variable x .

Sample Input 2:

$2*(x+(x+3)*9)$

2 6

Sample Output 2:

1

Explanation 2:

Arithmetic expression, A : $2*(x+(x+3)*9)$

P : 2 M : 6

Here, by solving the given equation, we will get:

If $x = 0$, then the remainder of dividing $2 * (x + (x+3) * 9)$ with 6 is 0, which is not equal to P .

If $x = 1$, then the remainder of division $2 * (x + (x+3) * 9)$ with 6 is 2, which is equal to P .

Hence, as an output, it will print 1 as the smallest value of variable x .